Supplemental Appendix

A Khan and Thomas (2008) Model

This appendix evaluates the precomputation method of Section 3 in the model of Khan and Thomas (2008). Unlike the Bloom et al. (2018) implementation, the price search in this exercise does not interpolate excess demand. The comparison between the forecast-deviation and price-level formulations therefore shows directly that centering the price grid on the forecast can substantially affect accuracy even when both formulations use the same number of price nodes.

A.1 Model

The Khan and Thomas (2008) model used here contains aggregate and idiosyncratic productivity shocks but no stochastic-volatility uncertainty shock. The household block is the same as in the baseline Bloom et al. (2018) model, so prices satisfy (11). Firms produce according to $y = z\varepsilon k^{\alpha} n^{\nu}$. Labor is chosen frictionlessly each period, whereas changing capital from its depreciated level $(1-\delta)k$ requires paying the stochastic fixed cost $\xi\omega$, where ξ is drawn independently each period from a distribution G with support $[0, \bar{\xi}]$.[13] Let μ denote the distribution of firms over (ε, k). Measuring the firm value in units of household marginal utility, $V \equiv pv$, the Bellman equation is

$$V^1(\varepsilon, k, \xi; z, \mu) = \max_{n \in \mathbf{R}_+} (z\varepsilon k^{\alpha} n^{\nu} - \omega n + (1-\delta)k)p + \max\left\{ -\xi\omega p + \max_{k^* \in \mathbf{R}_+} R(\varepsilon, k^*; z, \mu), R(\varepsilon, (1-\delta)k; z, \mu) \right\} \tag{35}$$

$$R(\varepsilon, k'; z, \mu) \equiv -k'p + \beta \mathbb{E}_{\varepsilon, z, \mu}\left[V^0(\varepsilon', k'; z', \mu') \right],$$

$$V^0(\varepsilon, k; z, \mu) \equiv \int_0^{\bar{\xi}} V^1(\varepsilon, k, \xi; z, \mu) G(d\xi)$$

Forecasting rules,

$$\mu' = \Gamma_{\mu}(z, \mu), \quad p = \Gamma_p(z, \mu)$$

Here R is the continuation return in marginal-utility units. Note that the second maximization in (35) reflects the firm's choice between investing to move to the new capital level k^* or remaining at the depreciated level $(1-\delta)k$. Importantly, the choice of k^* depends only on (z, ε, μ) and **not** on the firm's individual capital k.[14] The threshold level of the adjustment cost ξ that determines

[13] For comparability with Terry (2017), I omit the zero-investment band in the original Khan and Thomas (2008) specification.

[14] Since the adjustment cost $\xi\omega p$ is independent of k, the optimal choice of k^* is also independent of it.

Table 5: Computation time in the Khan–Thomas model

Model	Method	Total	VFI	Precompute policy	Simulation
KT	Baseline	16 min	3 sec	–	180 sec
KT	Proposed	80 sec	3 sec	3 sec	10 sec

whether the firm invests is given by:

$$\begin{aligned} \widehat{\xi}(\varepsilon, k; z, \mu) &= \frac{R(\varepsilon, k^*; z, \mu) - R(\varepsilon, (1-\delta)k; z, \mu)}{p\omega}, \\ \xi^*(\varepsilon, k; z, \mu) &= \min\{\bar{\xi}, \max\{0, \widehat{\xi}(\varepsilon, k; z, \mu)\}\}. \end{aligned} \tag{36}$$

The cutoff is restricted to the support of the cost distribution. For a continuous G, the choice at the cutoff does not affect aggregate quantities.

A.2 Precomputation: Computation Time and Accuracy

Speed comparison Table 5 reports the total computation time and its main components: the value-function iteration (VFI), the policy precomputation step, and the simulation step.

The proposed method reduces total computation time from 16 minutes to 80 seconds, corresponding to roughly a twelve-fold speedup. The value-function iteration itself takes 3 seconds in both methods, so the improvement does not come from solving the Bellman equation faster. Instead, the main driver of the speed gain is the simulation step. [15]

Accuracy check. I next examine whether this acceleration preserves the simulated dynamics of the baseline KS method. Figure 7 compares the simulation paths generated by the baseline KS method, the forecast-deviation formulation, and the price-level formulation over periods 1500–1600. The forecast-deviation series is almost indistinguishable from the baseline KS path, indicating that the proposed formulation reproduces the benchmark dynamics with essentially no visible discrepancy. By contrast, the price-level formulation displays noticeable deviations from the baseline path.

I further compare accuracy using investment statistics computed from the simulated paths. Table 6 reports moments of the investment rate i/k for the baseline KS method, the forecast-deviation formulation, and the price-level formulation. The investment statistics reinforce the evidence from the simulated paths. The forecast-deviation formulation matches the KS benchmark almost exactly across all reported moments. The mean and standard deviation of i/k are virtually unchanged, and the probabilities of inaction, positive investment, and negative investment remain extremely close to their KS values. By contrast, the price-level formulation generates

[15] In terms of computation time, there is essentially no difference between the price-level formulation and the forecast-deviation formulation; both formulations deliver the same speed improvement because they eliminate the repeated maximization inside the price-search loop.

Figure 7: Simulation path: baseline KS, forecast-deviation, and price-level formulations

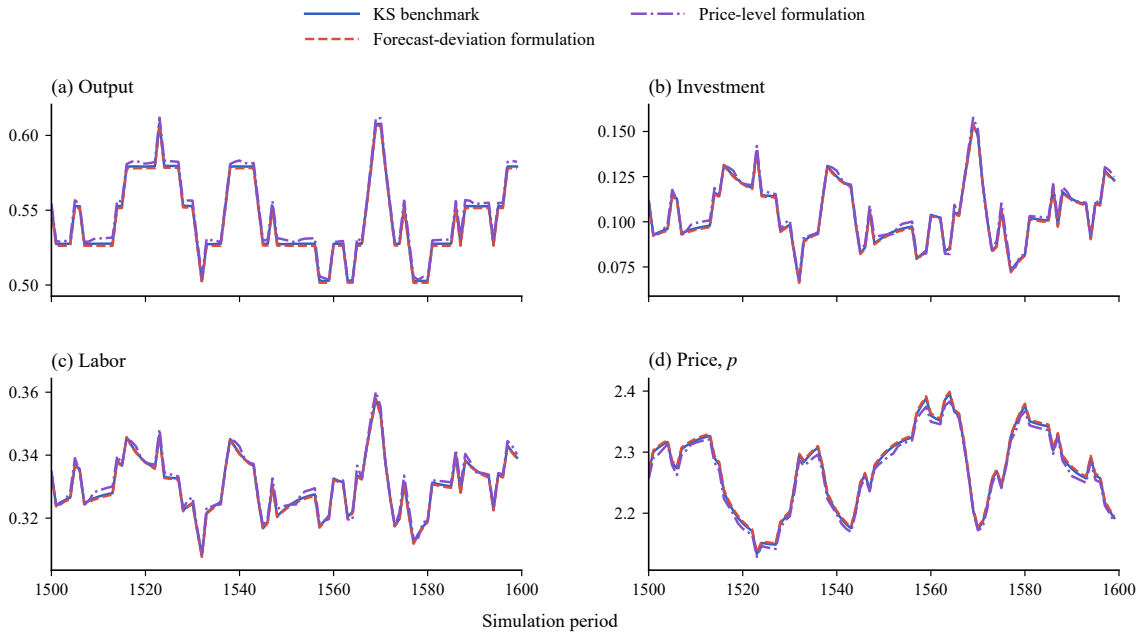


Table 6: Investment statistics in the Khan–Thomas model

Moment	KS	Forecast-deviation formulation	Price-level formulation
Mean i/k	0.09487	0.09488	0.09515
Std. i/k	0.25973	0.25982	0.25625
$\Pr(\text{inaction})$	0.76882	0.76764	**0.64339**
$\Pr(i/k \geq 0.2)$	0.17271	0.17266	**0.18483**
$\Pr(i/k \leq -0.2)$	0.02806	0.02804	**0.04364**
$\Pr(i/k > 0)$	0.18948	0.19015	**0.27216**
$\Pr(i/k < 0)$	0.04171	0.04221	**0.08445**

sizable discrepancies, especially along the extensive margin. The inaction probability falls from 0.76882 in the KS benchmark to 0.64339 under the price-level formulation, while the probabilities of positive and negative investment rise substantially.

B Additional Results of the NN-KS method for Bloom et al. (2018) Model

B.1 Hyperparameters and numerical settings

Table 7: NN–KS hyperparameters and numerical settings

Component	Setting	Value
Value network	Inputs; output	$(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K)$; scalar $\tilde{V}_{\text{nn}}$
	Hidden layers; activation	$256 \times 256 \times 256$; ReLU
Discrete gate	Inputs; output	seven states plus p_{error}; four regime logits
	Hidden layers; activation	$256 \times 256 \times 256$; ReLU
Conditional policy	Inputs; output	seven states, p_{error}, two-bit r; (k', n)
	Hidden layers; activation	$256 \times 256 \times 256$; ReLU
Optimization	Optimizer; learning-rate schedule	Adam; 10^{-4} toward 5×10^{-5}
	Main samples per component	500,000
	Main batch size; epochs	256; value 2, policy 1, gate 1
	Target-network update	$\tau = 0.05$
State grids	$(z, \varepsilon, k, n_{-1}, \sigma, \sigma_{-1}, K)$	$(5, 5, 91, 37, 2, 2, 10)$; 3,367,000 states
Price error	Input normalization	$p_{\text{error}}/0.05$ before network evaluation
	Ordinary additive perturbation	$[-0.05, 0.05]$
Simulation	Length; burn-in; price iterations	5,000; 500; at most 20
Precision	Networks; aggregation/distribution	FP32; FP64

B.2 Results

B.2.1 Bellman Errors: KS and NN–KS

This subsection reports relative Bellman errors for the saved KS and NN–KS value functions. Let $x_f = (k, n_{-1}, \varepsilon)$, $a = (k', n)$, and $k_0 = (1 - \delta_k)k$. Let $\tilde{V}_m(x_f; Z, K)$ denote the saved value function for method $m \in \{\text{KS}, \text{NN–KS}\}$, and let θ_m denote its associated forecasting-rule parameters. Define

$$(T_{\theta_m}\tilde{V}_m)(x_f; Z, K) = \max_{a \in \mathcal{A}(x_f, Z)} \Big\{ \widehat{p}_m \big[y - \widehat{w}_m n - (k' - k_0) - \text{AC}^k - \text{AC}^n(\widehat{w}_m) \big] + \beta \mathbb{E}\big[\tilde{V}_m\big(k', n, \varepsilon'; Z', \Gamma_K(Z, K; \theta_m)\big) \big] \Big\}, \tag{37}$$

where $\widehat{p}_m = \Gamma_p(Z, K; \theta_m)$ and $\widehat{w}_m = \chi / \widehat{p}_m$, as in (17). This is the same firm problem as in (18), evaluated using each method's value function and forecasting rules of the final outer loop.

I evaluate the Bellman operator itself rather than substituting the saved policy. The KS solution is computed on a discrete choice grid, so multilinear interpolation of $\tilde{V}_m$ is used to evaluate the continuation value at continuous choices. The NN–KS value network is already continuous in (k', n). For both methods, I maximize the right-hand side over $\mathcal{A}(x_f, Z)$ and use the resulting maximized value to construct the residual. The expectation is taken over all next-period shock realizations. The relative Bellman error is

$$B_m(x_f; Z, K) = \frac{\big|(T_{\theta_m}\tilde{V}_m)(x_f; Z, K) - \tilde{V}_m(x_f; Z, K)\big|}{\max\big\{\big|\tilde{V}_m(x_f; Z, K)\big|, 10^{-10}\big\}}. \tag{38}$$

I compute the mean of $B_m(x_f; Z, K)$ at 1,000 common grid nodes and 1,000 common off-grid states.

Table 8: Mean relative Bellman errors in the baseline model

Evaluation states	KS	NN–KS
Grid nodes	3.68667×10^{-7}	3.21681×10^{-4}
Off-grid states	5.44973×10^{-6}	3.22607×10^{-4}

Notes: Entries report $B_m(x_f; Z, K)$ as a ratio, not a percentage. Each evaluation sample contains 1,000 states shared by both methods. The NN–KS entries average the mean errors from five independent runs.

The NN–KS mean error is approximately 3.22×10^{-4}, or 0.032%, at both grid and off-grid states.

B.2.2 Shock-specific forecasting accuracy

Table 9 conditions forecasting accuracy on the aggregate state (z, σ, σ_{-1}).

Table 9: Bloom shock-specific forecast accuracy

Aggregate state	Price p				Capital K'			
(z, σ, σ_{-1})	KS		NN–KS		KS		NN–KS	
	RMSE (%)	R^2	RMSE (%)	R^2	RMSE (%)	R^2	RMSE (%)	R^2
				Den Haan statistics				
Maximum	3.80		3.59		5.68		5.77	
Mean	0.96		0.82		1.83		1.81	
(1,0,0)	0.61	0.76	0.58	0.82	0.33	0.98	0.37	0.99
(1,0,1)	0.10	0.99	0.20	0.96	0.05	1.00	0.12	1.00
(1,1,0)	0.19	0.97	0.23	0.96	0.52	0.95	0.59	0.94
(1,1,1)	0.29	0.91	0.28	0.95	0.11	1.00	0.15	1.00
(2,0,0)	0.51	0.81	0.50	0.86	0.34	0.98	0.35	0.99
(2,0,1)	0.13	0.98	0.14	0.99	0.08	1.00	0.08	1.00
(2,1,0)	0.39	0.84	0.41	0.84	0.27	0.97	0.27	0.98
(2,1,1)	0.33	0.93	0.38	0.94	0.11	1.00	0.16	1.00
(3,0,0)	0.43	0.92	0.46	0.92	0.28	0.99	0.27	0.99
(3,0,1)	0.19	0.99	0.23	0.98	0.10	1.00	0.11	1.00
(3,1,0)	0.23	0.98	0.29	0.97	0.37	0.98	0.45	0.99
(3,1,1)	0.37	0.95	0.41	0.93	0.12	1.00	0.14	1.00
(4,0,0)	0.49	0.93	0.54	0.86	0.32	0.99	0.31	0.99
(4,0,1)	0.17	0.99	0.22	0.97	0.09	1.00	0.12	1.00
(4,1,0)	0.46	0.95	0.31	0.93	0.39	0.98	0.40	0.98
(4,1,1)	0.36	0.94	0.44	0.88	0.12	1.00	0.13	1.00
(5,0,0)	0.53	0.91	0.51	0.89	0.33	0.99	0.30	0.99
(5,0,1)	0.11	0.99	0.22	0.96	0.22	0.99	0.16	1.00
(5,1,0)	0.33	0.98	0.34	0.93	0.13	1.00	0.12	1.00
(5,1,1)	0.31	0.97	0.31	0.93	0.12	1.00	0.12	1.00

Note: The table reports statistics conditional on $(z, \sigma, \sigma_{t-1})$. RMSE and Den Haan entries are percentage points of log deviations (log errors multiplied by 100).

B.2.3 Forecasting accuracy with finite-Frisch preferences

Table 10 summarizes the global accuracy of the price, wage, and aggregate capital forecasting rules. The price forecast has a lower RMSE and a higher R^2 than the baseline NN–KS price forecast in Table 4 (0.323% versus 0.46%, and 0.9878 versus 0.9747). The wage and aggregate capital forecasts also have low RMSEs and high R^2 values. Table 11 reports the corresponding statistics by aggregate state.

Table 10: Global forecasting accuracy with finite-Frisch preferences

Forecast	RMSE (%)	R^2	DH mean absolute error (%)	DH maximum absolute error (%)
Price p	0.323	0.9878	0.465	2.282
Wage w	0.331	0.9885	0.560	2.627
K′	0.082	0.9997	0.964	3.818

Notes: RMSE and Den Haan (DH) absolute errors are log errors multiplied by 100. R^2 is computed from the log forecast regression.

Table 11: State-specific forecasting accuracy with finite-Frisch preferences

Aggregate state	Price p		Wage w		Capital K'	
(z, σ, σ_{-1})	RMSE (%)	R^2	RMSE (%)	R^2	RMSE (%)	R^2
(1,0,0)	0.280	0.9705	0.407	0.9280	0.082	0.9996
(1,0,1)	0.155	0.9914	0.642	0.7501	0.036	0.9999
(1,1,0)	0.481	0.9141	0.584	0.8489	0.053	0.9998
(1,1,1)	0.428	0.9585	0.412	0.9486	0.057	0.9998
(2,0,0)	0.352	0.9520	0.422	0.9239	0.092	0.9994
(2,0,1)	0.184	0.9894	0.273	0.9640	0.056	0.9998
(2,1,0)	0.449	0.9253	0.524	0.8930	0.056	0.9997
(2,1,1)	0.329	0.9790	0.290	0.9763	0.051	0.9999
(3,0,0)	0.321	0.9633	0.323	0.9542	0.093	0.9993
(3,0,1)	0.372	0.9548	0.180	0.9841	0.039	0.9999
(3,1,0)	0.327	0.9753	0.369	0.9662	0.086	0.9994
(3,1,1)	0.288	0.9822	0.294	0.9724	0.046	0.9999
(4,0,0)	0.284	0.9739	0.259	0.9679	0.083	0.9995
(4,0,1)	0.298	0.9790	0.248	0.9746	0.044	0.9999
(4,1,0)	0.284	0.9798	0.347	0.9689	0.085	0.9994
(4,1,1)	0.279	0.9831	0.263	0.9772	0.051	0.9998
(5,0,0)	0.397	0.9417	0.253	0.9657	0.092	0.9994
(5,0,1)	0.570	0.8504	0.307	0.9194	0.041	0.9998
(5,1,0)	0.190	0.9920	0.320	0.9781	0.061	0.9998
(5,1,1)	0.312	0.9763	0.253	0.9798	0.058	0.9997

Notes: The tuple reports aggregate TFP z, current uncertainty state σ, and its lag σ_{-1}; z is indexed from 1 to 5. RMSE entries are log errors multiplied by 100, and R^2 is from the log forecast regression.

C Model Details and Additional Results for Section 5

This appendix describes how the household-preference extension in Section 5 is solved with NN–KS and provides additional quantitative results. It then discusses the calibration of the Frisch

elasticity, examines the role of labor adjustment costs, and varies consumption–hours complementarity.

C.1 Model and NN–KS Implementation

The firm side of the model is unchanged from Section 2. A firm has the individual state $x_f = (k, n_{-1}, \varepsilon)$, produces $y = z\varepsilon k^{\alpha} n^{\nu}$, and chooses next-period capital k' and current employment n subject to the capital and labor adjustment costs described in Section 2.1. The preference extension replaces the baseline household utility function with a general period utility $U(C, N)$. The two household price conditions are

$$p = U_C(C, N), \qquad w = -\frac{U_N(C, N)}{U_C(C, N)}. \tag{39}$$

Unlike in the baseline model, p and w cannot generally be recovered from one another. Consequently, the simulation searches jointly for both prices. For each candidate pair (p, w), the firm policies imply aggregate employment $N(p, w)$ and, through the resource constraint, consumption $C(p, w)$. The two market-clearing residuals are

$$p - U_C\big(C(p,w), N(p,w)\big) = 0, \qquad w + \frac{U_N\big(C(p,w), N(p,w)\big)}{U_C\big(C(p,w), N(p,w)\big)} = 0. \tag{40}$$

To approximate the distributional state, I retain aggregate capital K_t and add lagged aggregate employment N_{t-1} as a second moment. The latter is useful to forecast the wage. With $Z_t = (z_t, \sigma_t, \sigma_{t-1})$, NN–KS therefore uses three forecasting rules:

$$K_{t+1} = \Gamma_K(Z_t, K_t, N_{t-1}; \theta), \tag{41}$$

$$\widehat{p}_t = \Gamma_p(Z_t, K_t, N_{t-1}; \theta), \tag{42}$$

$$\widehat{w}_t = \Gamma_w(Z_t, K_t, N_{t-1}; \theta). \tag{43}$$

To evaluate the continuation value, current aggregate employment N_t, which becomes the lagged-employment moment next period, is backed out from the household wage condition at $(\widehat{p}_t, \widehat{w}_t)$. For example, under the separable finite-Frisch specification,

$$p_t w_t = \chi N_t^{\phi} \quad \Longrightarrow \quad \widehat{N}_t = \left(\frac{\widehat{p}_t \widehat{w}_t}{\chi}\right)^{1/\phi}, \qquad \phi > 0. \tag{44}$$

For the other preference specifications, $\widehat{N}_t$ is obtained from the corresponding household price equations in (39). Thus, the continuation value at a candidate firm action uses

$$\widetilde{V}_{\theta}\Big(k', n, \varepsilon'; Z', \Gamma_K(Z_t, K_t, N_{t-1}; \theta), \widehat{N}_t\Big). \tag{45}$$

Relative to the implementation in Section 4, the value network therefore gains lagged aggregate employment as an input:

$$\widetilde{V}_{\text{nn}} = \widetilde{V}_{\text{nn}}(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K, N_{-1}). \tag{46}$$

The discrete and conditional policy networks additionally take the candidate wage. In the implementation, the two candidate prices are represented as additive deviations from their respective forecasts,

$$p = \Gamma_p(Z, K, N_{-1}; \theta) + p_{\text{error}}, \qquad w = \Gamma_w(Z, K, N_{-1}; \theta) + w_{\text{error}}. \tag{47}$$

Hence, their common economic inputs are

$$(k, n_{-1}, \varepsilon, z, \sigma, \sigma_{-1}, K, N_{-1}, p_{\text{error}}, w_{\text{error}}), \tag{48}$$

with the adjustment-regime indicators also supplied to the conditional policy network. During policy training, p_{error} and w_{error} are sampled independently from the interval $[-0.05, 0.05]$, even though current aggregate labor N for the continuation value is computed by the forecasting rule with $p_{\text{error}} = w_{\text{error}} = 0$. Value-network updates use the forecast pair, $p_{\text{error}} = w_{\text{error}} = 0$.

C.2 Calibration of the Frisch Elasticity

Section 5.3 and Figure 6 compare the original model with separable preferences at Frisch elasticities $1/\phi \in \{0.5, 1, 2\}$. The finite-elasticity cases use the common reference allocation described in Section 5.2. This appendix reports the unconditional labor moments that motivate choosing an elasticity of 2 for the preference comparison in Appendix C.4.

Calibration of the Frisch elasticity. Although a smaller elasticity strengthens the wage response that supports employment after an uncertainty shock, it also dampens unconditional labor fluctuations. Table 12 reports labor business-cycle statistics from the unconditional simulation of each case. The standard deviation of labor rises from 0.868% at an elasticity of 0.5 to 1.050% at 1 and 1.601% at 2. The labor-to-output volatility ratio increases from 0.665 to 0.747 and 0.989, and the correlation of labor with output also rises. The corresponding U.S. statistics reported by Bloom et al. (2018, Table 7) are 2.0%, 1.3, and 0.9. A larger elasticity therefore moves all three labor moments toward the data in this comparison, although the elasticity of 2 still understates them.

This calibration consideration is consistent with Bloom et al. (2018), who emphasize that their high assumed Frisch elasticity helps generate realistic hours volatility relative to output. I retain a finite elasticity to preserve the wage channel, while choosing a relatively elastic specification within the range supported by literature.

Table 12: Labor business-cycle statistics across Frisch elasticities

Specification	$\mathrm{sd}(N)$ (%)	$\mathrm{sd}(N)/\mathrm{sd}(Y)$	$\mathrm{corr}(N, Y)$
Frisch $= 0.5$	0.868	0.665	0.544
Frisch $= 1$	1.050	0.747	0.639
Frisch $= 2$	1.601	0.989	0.744
U.S. data	2.0	1.3	0.9

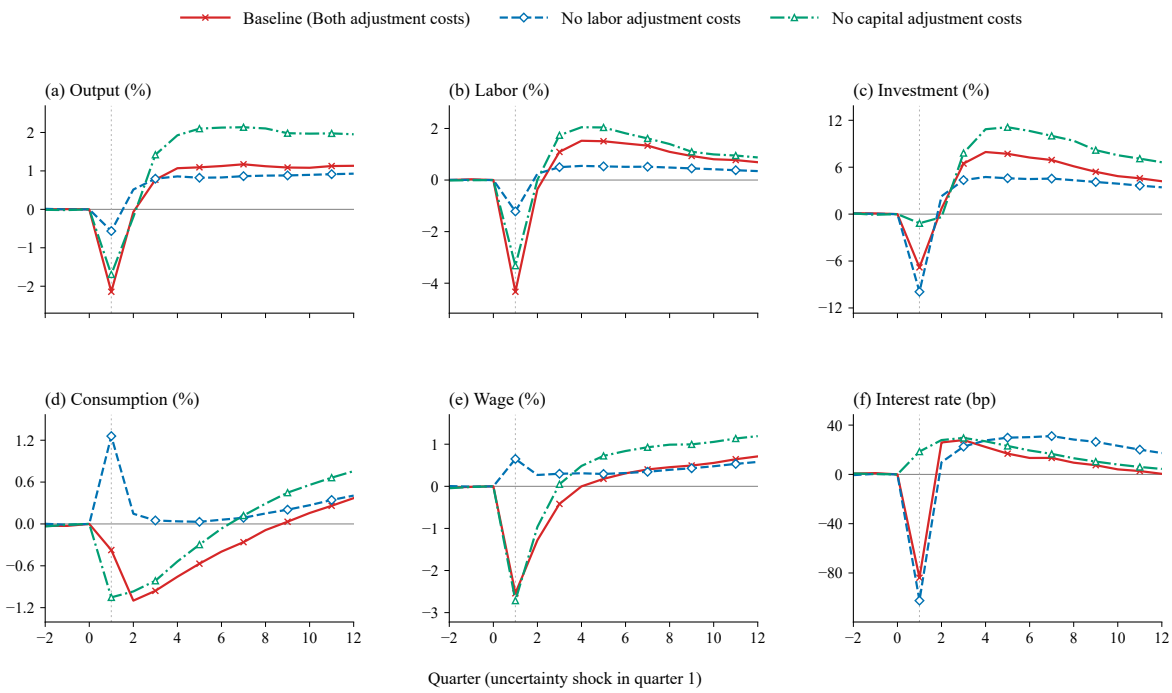
Notes: Model statistics use 5,000-quarter simulations, with the first 500 quarters discarded. Each series is expressed in logarithms and HP-filtered with $\lambda = 1600$; the cyclical component is multiplied by 100. Standard deviations use the sample definition. U.S. moments are from Bloom et al. (2018, Table 7), for 1972Q1–2010Q4; labor is total nonfarm business-sector hours. Data entries retain the precision reported in that source.

C.3 Labor Adjustment Costs and Finite Frisch Elasticity

The wage channel in Section 5.3 operates in an economy with labor adjustment costs. Figure 8 examines their role by removing labor adjustment costs or capital irreversibility while retaining separable preferences and a Frisch elasticity of 2. The labor counterfactual sets $F^L = H = 0$, removing both fixed and linear employment adjustment costs. The capital counterfactual sets $S = 0$; the fixed capital adjustment cost F^K is zero throughout. All other structural parameters are unchanged.

Without labor adjustment costs, employment falls by 1.22%, but consumption rises by 1.26% and the wage rises by 0.65%, despite the finite Frisch elasticity. Investment falls by 9.93%. Removing labor costs makes employment a static choice: current labor demand depends on current productivity, capital, and the wage, rather than on the value of preserving employment for future periods. The simultaneous decline in output, consumption, investment, and employment is therefore lost. Labor adjustment costs and the finite Frisch elasticity work jointly in the baseline: the costs make labor demand sensitive to uncertainty through the value of waiting, while the elasticity governs how adjustment is divided between wages and employment.

Figure 8: Uncertainty-shock responses with adjustment costs removed



Notes: The shock occurs in quarter 1. Panels (a)–(e) show $100 \log(\overline{X}_t / \overline{X}_0)$, using arithmetic mean levels across 2,000 economies per case. Panel (f) shows the change in the quarterly real risk-free rate in basis points, computed as $r_t^f = [\mathbb{E}_t(\beta p_{t+1}/p_t)]^{-1} - 1$; it is not annualized. All cases have separable preferences and Frisch elasticity 2 and share the same shock realizations. The counterfactuals remove labor adjustment costs ($F^L = H = 0$) or capital irreversibility ($S = 0$), with $F^K = 0$ throughout. Each path uses its own pre-shock mean.

By contrast, removing capital irreversibility preserves comovement. Output, employment, investment, and consumption fall by 1.69%, 3.32%, 1.17%, and 1.05%, respectively. The investment decline is much smaller than the baseline 6.81%. Capital irreversibility thus amplifies the investment contraction, while labor adjustment costs are central to the consumption–employment response in this calibration.

C.4 Consumption–Hours Complementarity

Bloom et al. (2018) discuss GHH preferences as a possible way to change the consumption response. Under GHH preferences, consumption and hours are complements, so a decline in employment reduces the marginal utility of a given amount of consumption. This subsection examines whether introducing this channel causes consumption to fall with employment and thereby generates comovement after an uncertainty shock.

To vary complementarity while controlling the curvature of marginal utility, I use the quasi-

separable preference family of Bilbiie et al. (2025):

$$U(C,N)=\frac{1}{1-\frac{\xi}{1-\gamma}}A(C,N)^{1-\frac{\xi}{1-\gamma}},\qquad A(C,N)\equiv\frac{C^{1-\gamma}}{1-\gamma}-\chi\frac{N^{1+\phi}}{1+\phi}. \tag{49}$$

The induced marginal-utility price and real wage are

$$p_t=U_C(C_t,N_t)=C_t^{-\gamma}A(C_t,N_t)^{-\frac{\xi}{1-\gamma}}, \tag{50}$$

$$w_t=-\frac{U_N(C_t,N_t)}{U_C(C_t,N_t)}=\chi C_t^{\gamma}N_t^{\phi}. \tag{51}$$

Following Bilbiie et al. (2025), let ζ denote the local curvature of marginal utility, κ consumption–hours complementarity, and γ the wealth effect on labor supply. For equation (49),

$$\kappa(C,N)=\frac{\xi}{1-\gamma}\frac{C^{1-\gamma}}{A(C,N)},\qquad \boxed{\zeta(C,N)=\kappa(C,N)+\gamma}. \tag{52}$$

Thus, holding ζ fixed while increasing complementarity requires a weaker wealth effect. Around the reference allocation, the two price equations can be written as

$$\hat{p}_t\simeq-\bar{\zeta}\hat{c}_t+\bar{\kappa}\omega\hat{n}_t,\qquad \hat{w}_t\simeq\gamma\hat{c}_t+\phi\hat{n}_t,\qquad \omega\equiv\frac{\bar{w}\bar{N}}{\bar{C}}, \tag{53}$$

where hats denote log deviations and bars denote values at the reference allocation. A decline in consumption raises p_t, whereas a decline in hours lowers it when $\kappa>0$. Through the wage equation, reducing γ also weakens the effect of consumption on labor supply.

I set $\bar{\zeta}=1$, as under log-separable preferences, and compare $\bar{\kappa}=0$, 0.5, and 1, with $\gamma=1-\bar{\kappa}$. The endpoints are separable and GHH preferences, while the intermediate case is quasi-separable. All three specifications use a Frisch elasticity of 2. Table 13 summarizes the calibration. Appendices D and D.1 provide the preference mapping and calibration details.

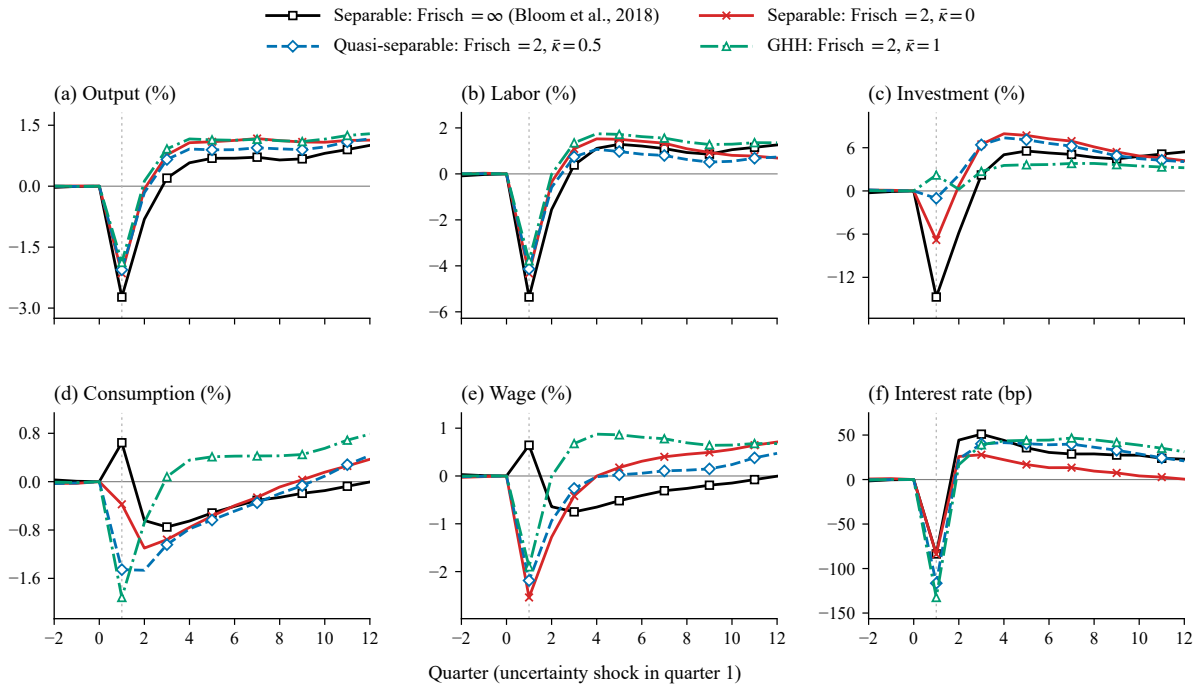
Table 13: Preference parameters in Figure 9

Specification	$\bar{\kappa}$	$\bar{\zeta}$	γ	Frisch elasticity
Original	0	1	1	∞
Separable	0	1	1	2
Quasi-separable	0.5	1	0.5	2
GHH	1	1	0	2

Notes: $\bar{\zeta}$, $\bar{\kappa}$, and the Frisch elasticity are evaluated at the reference allocation. In the quasi-separable family, ζ and κ can vary away from this allocation, while γ is constant.

The blue and green lines in Figure 9 show how consumption–hours complementarity changes the finite-Frisch responses. Output and employment still fall by roughly 2% and 4% on impact. Under the quasi-separable specification, consumption falls by 1.46%, but p_t rises by only 0.23%: the decline in hours offsets much of the marginal-utility increase caused by lower consumption.

Figure 9: Uncertainty-shock responses under alternative household preferences



Notes: The shock occurs in quarter 1. Panels (a)–(e) show $100\log(\overline{X}_t/\overline{X}_0)$, using 2,000 NN–KS economies per specification. Panel (f) shows the change in the quarterly real risk-free rate in basis points, computed as $r_t^f = [\mathbb{E}_t(\beta p_{t+1}/p_t)]^{-1} - 1$; it is not annualized. The infinite-Frisch benchmark corresponds to the original model in Figure 5. $\bar{\kappa}$ denotes consumption–hours complementarity at the reference allocation; all cases have $\bar{\zeta} = 1$. Each path uses its own pre-shock mean. Table 13 gives the calibration.

Investment falls by 1.03%, so output, consumption, investment, and employment continue to comove.

At the GHH endpoint, consumption falls by 1.92%, while p_t falls by 0.32%. The effect of lower hours on marginal utility now more than offsets the effect of lower consumption. The fall in p_t reduces the utility cost of current investment relative to future payoffs, supporting an impact investment increase of 2.21%. GHH preferences therefore reverse the positive consumption response of the original model, but investment rises rather than falling with the other aggregates.

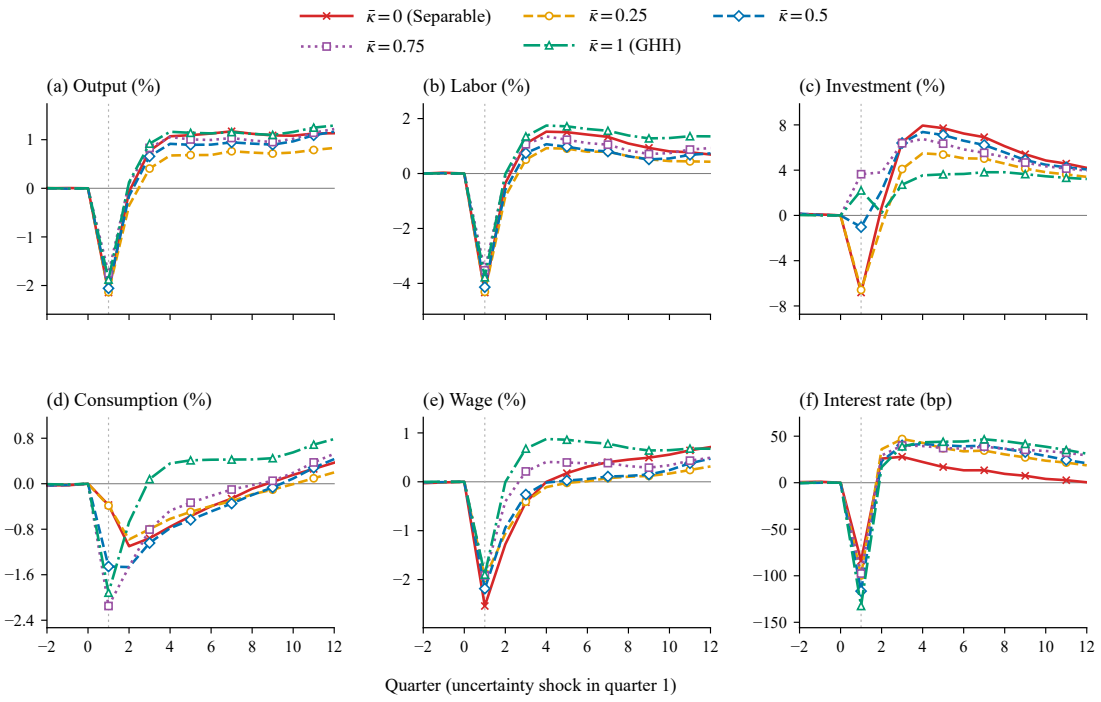
This effect on marginal utility also operates through the interest rate. A lower current p_t relative to its expected next-period value raises the expected stochastic discount factor $\beta\mathbb{E}_t[p_{t+1}]/p_t$ and lowers the real risk-free rate. Panel (f) shows impact rate declines of 83.9 basis points under finite-Frisch separability, 116.5 basis points under quasi-separability, and 132.5 basis points under GHH preferences. Lower interest rates make future payoffs more valuable relative to current expenditure and support investment. Appendix C.5 extends the comparison to five values of $\bar{\kappa}$.

C.5 Sensitivity to Consumption–Hours Complementarity

This subsection extends the comparison in Appendix C.4 to five values of consumption–hours complementarity, $\bar{\kappa} \in \{0, 0.25, 0.5, 0.75, 1\}$. All specifications target $\bar{\zeta} = 1$ and a Frisch elasticity of

2 at the reference allocation, using the calibration in Appendix D.1. The endpoints are separable and GHH preferences, and the three interior values use quasi-separable preferences. The identity $\bar{\zeta} = \bar{\kappa} + \gamma$ implies $\gamma = 1 - \bar{\kappa}$: increasing complementarity also weakens the wealth effect on labor supply. The exercise therefore traces a calibrated family of preferences, rather than varying complementarity while holding the wealth effect fixed.

Figure 10: Uncertainty-shock responses as consumption–hours complementarity varies



Notes: The shock occurs in quarter 1. Panels (a)–(e) show $100\log(\overline{X}_t/\overline{X}_0)$, using arithmetic mean levels across 2,000 economies per specification. Panel (f) shows the change in the quarterly real risk-free rate in basis points, computed as $r_t^f = [\mathbb{E}_t(\beta p_{t+1}/p_t)]^{-1} - 1$; it is not annualized. The time window and shared cases match Figure 9. All cases target $\bar{\zeta} = 1$ and a Frisch elasticity of 2. Each path uses its own pre-shock mean.

Figure 10 shows that output, hours, consumption, and the real wage fall on impact in all five cases. Investment is more sensitive to the preference specification: its impact response is negative at $\bar{\kappa} = 0$, 0.25, and 0.5, but positive at 0.75 and 1. Thus, the simultaneous decline in output, consumption, and investment found for the intermediate specification in Appendix C.4 remains a feature of part of this family, rather than a prediction for every degree of complementarity.

Non-monotonic responses. Table 14 makes clear that the responses are not ordered monotonically by $\bar{\kappa}$. For example, raising $\bar{\kappa}$ from 0.5 to 0.75 changes investment's impact response from -1.03% to 3.64%, but a further increase to 1 lowers it to 2.21%. Consumption falls by 1.46%, 2.15%, and 1.92% in these three cases. The contractions in output and hours also become larger between 0.75 and 1. Neither the wage response nor the response of marginal utility follows a monotonic ordering over the five calibrations.

Table 14: Impact responses across complementarity calibrations

$\bar{\kappa}$	γ	Output	Labor	Investment	Consumption	Wage	p
0	1	−2.14	−4.33	−6.81	−0.37	−2.54	0.39
0.25	0.75	−2.14	−4.31	−6.58	−0.39	−1.97	−0.25
0.5	0.5	−2.06	−4.14	−1.03	−1.46	−2.19	0.23
0.75	0.25	−1.75	−3.52	3.64	−2.15	−1.91	0.60
1	0	−1.88	−3.78	2.21	−1.92	−1.90	−0.32

Notes: Entries for quantities and prices are impact log deviations in percent from each specification's pre-shock cross-economy mean, as in Figure 10. The wealth-effect parameter is $\gamma = 1 - \bar{\kappa}$.

D A Unified Representation of Separable and GHH Preferences

This appendix relates the separable, GHH, and quasi-separable preferences used in the quantitative analysis through the framework of Bilbiie et al. (2025). The key advantage of this representation is that it separates three distinct properties of preferences: the curvature of the marginal utility of consumption, consumption–hours complementarity, and the wealth effect on labor supply.

Consider a general instantaneous utility function

$$U(C, N),$$

where C denotes consumption and N denotes hours worked. Following Bilbiie et al. (2025), define the local curvature of the marginal utility of consumption as

$$\zeta(C, N) \equiv -\left.\frac{\partial \log U_C}{\partial \log C}\right|_N = -\frac{C U_{CC}}{U_C}. \tag{54}$$

I measure consumption–hours complementarity by

$$\kappa(C, N) \equiv -\left.\frac{\partial \log(-U_N)}{\partial \log C}\right|_N = -\frac{C U_{CN}}{U_N}. \tag{55}$$

Because $U_N < 0$, positive Edgeworth complementarity, $U_{CN} > 0$, implies $\kappa > 0$. Finally, define the strength of the wealth effect on labor supply by the response of the marginal rate of substitution to consumption:

$$\gamma(C, N) \equiv \left.\frac{\partial \log\left(-U_N / U_C\right)}{\partial \log C}\right|_N. \tag{56}$$

Combining (54)–(56) gives

$$\begin{aligned} \gamma(C, N) &= \frac{C U_{CN}}{U_N} - \frac{C U_{CC}}{U_C} \\ &= \zeta(C, N) - \kappa(C, N), \end{aligned} \tag{57}$$

or, equivalently,

$$\boxed{\zeta(C,N) = \kappa(C,N) + \gamma(C,N).} \tag{58}$$

This decomposition is useful because it makes clear that the curvature of marginal utility need not be identified with either consumption–hours complementarity or the wealth effect individually. Instead, ζ reflects both forces.

Separable preferences. Consider first the standard separable specification

$$U^{\text{SEP}}(C,N) = \frac{C^{1-\eta}}{1-\eta} - \chi\frac{N^{1+\phi}}{1+\phi}. \tag{59}$$

Since $U_{CN} = 0$, separability implies

$$\kappa^{\text{SEP}} = 0. \tag{60}$$

Moreover,

$$-\frac{CU_{CC}}{U_C} = \eta,$$

so that

$$\zeta^{\text{SEP}} = \gamma^{\text{SEP}} = \eta. \tag{61}$$

Thus, under separable preferences, all of the curvature in the marginal utility of consumption appears as a wealth effect on labor supply. There is no consumption–hours complementarity.

GHH preferences. At the opposite endpoint, consider GHH preferences,

$$U^{\text{GHH}}(C,N) = \frac{X(C,N)^{1-\eta}}{1-\eta}, \qquad X(C,N) \equiv C - \chi\frac{N^{1+\phi}}{1+\phi}. \tag{62}$$

The marginal utilities are

$$U_C = X^{-\eta}, \qquad U_N = -\chi N^{\phi} X^{-\eta}.$$

It follows that

$$\zeta^{\text{GHH}}(C,N) = \eta\frac{C}{X(C,N)} \tag{63}$$

and

$$\kappa^{\text{GHH}}(C,N) = \eta\frac{C}{X(C,N)}. \tag{64}$$

Hence,

$$\boxed{\zeta^{\text{GHH}} = \kappa^{\text{GHH}}, \qquad \gamma^{\text{GHH}} = 0.} \tag{65}$$

GHH preferences therefore represent the other endpoint: all of the curvature of the marginal utility

of consumption is associated with consumption–hours complementarity, while the wealth effect on labor supply is eliminated completely.

Notice that the primitive curvature parameter η in the GHH utility function is not itself equal to ζ. Rather, the relevant local curvature is

$$\zeta^{\text{GHH}}(C,N) = \eta \frac{C}{X(C,N)}.$$

This distinction is important when comparing GHH and separable preferences while holding ζ fixed.

Quasi-separable GHH–CRRA preferences. The quasi-separable specification of Bilbiie et al. (2025) nests these two polar cases while allowing complementarity and the wealth effect to be controlled separately:

$$U^{\text{QS}}(C,N) = \frac{1}{1-\frac{\xi}{1-\gamma}} \left[\frac{C^{1-\gamma}}{1-\gamma} - \chi \frac{N^{1+\phi}}{1+\phi} \right]^{1-\frac{\xi}{1-\gamma}}. \tag{66}$$

Define

$$A(C,N) \equiv \frac{C^{1-\gamma}}{1-\gamma} - \chi \frac{N^{1+\phi}}{1+\phi}, \qquad a \equiv \frac{\xi}{1-\gamma}. \tag{67}$$

The marginal utilities are

$$U_C = C^{-\gamma} A^{-a}, \qquad U_N = -\chi N^{\phi} A^{-a}. \tag{68}$$

Therefore,

$$\zeta^{\text{QS}}(C,N) = \gamma + a \frac{C^{1-\gamma}}{A(C,N)}, \tag{69}$$

$$\kappa^{\text{QS}}(C,N) = a \frac{C^{1-\gamma}}{A(C,N)} = \frac{\xi}{1-\gamma} \frac{C^{1-\gamma}}{A(C,N)}. \tag{70}$$

Consequently,

$$\boxed{\zeta^{\text{QS}}(C,N) = \kappa^{\text{QS}}(C,N) + \gamma.} \tag{71}$$

The two benchmark specifications arise as limiting cases. When complementarity disappears, $\kappa = 0$, the quasi-separable specification converges to separable CRRA preferences. At the other endpoint, setting $\gamma = 0$ eliminates the wealth effect and gives

$$A(C,N) = C - \chi \frac{N^{1+\phi}}{1+\phi},$$

so that (66) reduces exactly to the GHH form.

D.1 Calibration with a Common Consumption Curvature

To isolate the role of consumption–hours complementarity, I calibrate the quasi-separable preference family of Bilbiie et al. (2025) while holding the curvature of marginal utility and the Frisch elasticity of labor supply fixed at a common reference allocation. Specifically, I impose

$$\bar{\zeta} = 1, \qquad \bar{\varepsilon}_F = 2, \tag{72}$$

where bars denote evaluation at the reference steady state and ε_F denotes the Frisch elasticity.

The common reference values are

$$(\bar{C}, \bar{N}, \bar{w}) = (0.73,\ 0.299,\ 1.451), \tag{73}$$

which imply

$$\omega \equiv \frac{\bar{w}\bar{N}}{\bar{C}} \simeq 0.5943. \tag{74}$$

I vary the target complementarity parameter over $\bar{\kappa} \in \{0, 0.25, 0.5, 0.75, 1\}$ and recalibrate the four primitive preference parameters $(\gamma, \phi, \chi, \xi)$ accordingly. The calibration proceeds in four steps.

Step 1: Wealth-effect parameter γ. Following Bilbiie et al. (2025), the curvature of marginal utility, consumption–hours complementarity, and the wealth effect on labor supply satisfy

$$\zeta(C, N) = \kappa(C, N) + \gamma. \tag{75}$$

Since I impose $\bar{\zeta} = 1$ at the reference allocation, the wealth-effect parameter is determined by

$$\boxed{\gamma = \bar{\zeta} - \bar{\kappa} = 1 - \bar{\kappa}.} \tag{76}$$

Thus, increasing complementarity while holding the curvature of marginal utility fixed necessarily reduces the wealth effect on labor supply. The endpoints $\bar{\kappa} = 0$ and $\bar{\kappa} = 1$ correspond to separable and GHH preferences, respectively.

Step 2: Labor curvature parameter ϕ. Under quasi-separable preferences, the Frisch elasticity does not generally equal $1/\phi$. The Frisch elasticity measures the response of labor supply to the real wage while holding the marginal utility of consumption, $p = U_C(C, N)$, constant.

Using the local household optimality conditions in equation (53), the log differentials around the reference allocation satisfy

$$d \log p = -\bar{\zeta}\, d \log C + \bar{\kappa}\omega\, d \log N, \tag{77}$$

$$d \log w = \gamma\, d \log C + \phi\, d \log N. \tag{78}$$

Holding marginal utility constant, $d \log p = 0$, implies

$$d \log C = \frac{\bar{\kappa}\omega}{\bar{\zeta}} d \log N. \tag{79}$$

Substituting this expression into the wage equation yields

$$\boxed{\bar{\varepsilon}_F^{-1} = \left. \frac{d \log w}{d \log N} \right|_p = \phi + \frac{\gamma \bar{\kappa}\omega}{\bar{\zeta}}.} \tag{80}$$

To maintain a common Frisch elasticity of 2, I therefore calibrate

$$\boxed{\phi = \frac{1}{\bar{\varepsilon}_F} - \frac{\gamma \bar{\kappa}\omega}{\bar{\zeta}} = \frac{1}{2} - \gamma \bar{\kappa}\omega.} \tag{81}$$

At the separable endpoint, $\bar{\kappa} = 0$, and at the GHH endpoint, $\gamma = 0$, this reduces to $\phi = 1/2$. For interior quasi-separable specifications, ϕ is adjusted to offset the contribution of complementarity to the Frisch elasticity.

Step 3: Labor-disutility parameter χ. I next choose the scale of labor disutility to preserve the household's intratemporal optimality condition at the common reference allocation.

For the quasi-separable preference family, the real wage satisfies

$$w = -\frac{U_N(C, N)}{U_C(C, N)} = \chi C^{\gamma} N^{\phi}. \tag{82}$$

Evaluating this condition at $(\bar{C}, \bar{N}, \bar{w})$ gives

$$\boxed{\chi = \frac{\bar{w}}{\bar{C}^{\gamma} \bar{N}^{\phi}}.} \tag{83}$$

Thus, changes in γ and ϕ are accompanied by changes in χ so that the same reference consumption, employment, and real wage continue to satisfy the household's intratemporal condition.

Step 4: Outer curvature parameter ξ. Finally, I choose ξ to attain the target degree of consumption–hours complementarity at the reference allocation.

For the quasi-separable specification, define

$$A(C, N) = \frac{C^{1-\gamma}}{1-\gamma} - \chi \frac{N^{1+\phi}}{1+\phi}. \tag{84}$$

As shown in equation (70), complementarity satisfies

$$\kappa(C,N) = \frac{\xi}{1-\gamma}\frac{C^{1-\gamma}}{A(C,N)}. \tag{85}$$

For $\gamma < 1$, let

$$\bar{A} = \frac{\bar{C}^{1-\gamma}}{1-\gamma} - \chi\frac{\bar{N}^{1+\phi}}{1+\phi}. \tag{86}$$

Imposing $\kappa(\bar{C},\bar{N}) = \bar{\kappa}$ gives

$$\boxed{\xi = \bar{\kappa}(1-\gamma)\frac{\bar{A}}{\bar{C}^{1-\gamma}}.} \tag{87}$$

Using the calibration of χ in equation (83), the reference aggregator can be rewritten as

$$\bar{A} = \bar{C}^{1-\gamma}\left[\frac{1}{1-\gamma} - \frac{\omega}{1+\phi}\right]. \tag{88}$$

Substituting this expression into equation (87) and using $1-\gamma = \bar{\kappa}$ yields

$$\boxed{\xi = \bar{\kappa}\left(1 - \frac{\bar{\kappa}\omega}{1+\phi}\right).} \tag{89}$$

Therefore, conditional on the target $\bar{\kappa}$ and the common reference allocation, all four primitive preference parameters are determined analytically.

At the separable endpoint, $\bar{\kappa} = 0$ and $\gamma = 1$, I use the limiting log-separable utility directly:

$$U^{\text{SEP}}(C,N) = \log C - \chi\frac{N^{1+\phi}}{1+\phi}. \tag{90}$$

The parameter ξ is not separately required in this representation.

At the GHH endpoint, $\bar{\kappa} = 1$ and $\gamma = 0$, the aggregator becomes

$$X(C,N) = C - \chi\frac{N^{1+\phi}}{1+\phi}. \tag{91}$$

Equation (89) then gives

$$\boxed{\xi^{\text{GHH}} = \frac{\bar{X}}{\bar{C}} = 1 - \frac{\omega}{1+\phi}.} \tag{92}$$

At this endpoint, ξ coincides with the GHH curvature parameter η introduced in equation (62).

Calibrated preference parameters. Table 15 reports the resulting primitive parameters for the five complementarity specifications. The numerical values are obtained by applying the preceding calibration equations to the common reference allocation.

The calibration progressively replaces the wealth-effect component of marginal-utility cur-

Table 15: Calibration of primitive preference parameters

Specification	$\bar{\kappa}$	γ	ϕ	χ	ξ
Separable	0	1.00	0.5000	3.6350	–
Quasi-separable	0.25	0.75	0.3886	2.9371	0.2232
Quasi-separable	0.50	0.50	0.3514	2.5958	0.3901
Quasi-separable	0.75	0.25	0.3886	2.5094	0.5092
GHH	1.00	0.00	0.5000	2.6536	0.6038

Notes: All specifications target $\bar{\zeta} = 1$ and $\bar{\varepsilon}_F = 2$ at the common reference allocation $(\bar{C}, \bar{N}, \bar{w}) = (0.73, 0.299, 1.451)$. The primitive parameters are calibrated using equations (76), (81), (83), and (89). The separable specification is defined directly using log utility and does not require ξ. The original infinite-Frisch benchmark is not included in this calibration.

vature with consumption–hours complementarity, while keeping both the local curvature of marginal utility and the local Frisch elasticity fixed. In particular, the adjustment of ϕ ensures that differences across preference specifications are not driven by differences in the Frisch elasticity at the reference allocation.

These restrictions are imposed locally. Under interior quasi-separable preferences, both $\zeta(C, N)$ and $\kappa(C, N)$ generally vary with consumption and employment away from the reference allocation, although γ remains constant. Consequently, the Frisch elasticity

$$\varepsilon_F(C, N)^{-1} = \phi + \frac{\gamma\kappa(C, N)}{\zeta(C, N)} \frac{w(C, N)N}{C} \tag{93}$$

can also vary along the equilibrium path. At the separable and GHH endpoints, however, the Frisch elasticity remains equal to $1/\phi = 2$ globally.